PROJECT PROPOSAL

Riayah Plus — HealthOS Platform

A Vertically Integrated Healthcare Operating System

Budget	Timeline
$3,000 USD	6 Weeks

Submitted By	Stack
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Revision note: This version removes all AI-assisted modules from the original proposal, adds local Qatari bank payment integration, and adds GPS-based location tracking for support/field workers.

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1. Project Overview

Riayah Plus is a vertically integrated, technology-enabled healthcare provider built as a Healthcare Operating System (HealthOS). The platform digitizes the full patient lifecycle — from initial onboarding through clinical assessments, care planning, service scheduling, delivery, and outcomes tracking — with full audit compliance and role-based access control.

This proposal outlines the scope, technical architecture, milestone breakdown, and investment plan required to deliver a production-grade platform across web and mobile within 6 weeks.

1.1 Objectives

- Deliver a multi-role web and mobile platform covering all core clinical workflows
- Implement structured care episode lifecycle management end-to-end
- Support telehealth as a primary care delivery channel
- Integrate secure local payment processing with Qatari banks for invoicing and quotations
- Enable GPS-based location tracking for support workers during care delivery visits
- Ensure full audit trail, HIPAA-aligned data handling, and regional data residency
- Build for scalability: multi-clinic, multi-department, multi-tenant ready architecture

1.2 Technology Stack

Layer	Technology
Frontend (Web)	React.js (Next.js), Tailwind CSS, Redux Toolkit, React Query
Mobile App	React Native (Expo), NativeWind, React Navigation, Zustand
Backend API	Node.js, Express.js, REST API with modular domain routing
Database	PostgreSQL with Prisma ORM, structured schema with full relational integrity
Authentication	JWT + Refresh Token rotation, RBAC middleware, bcrypt password hashing
Telehealth	Daily.co or Twilio Video API integration with waiting room & session recording

Layer	Technology
Payments	Local Qatari payment gateway (e.g. PayTabs / Fatora / Skipcash) supporting NAPS-linked debit cards and major Qatari bank cards (QNB, Doha Bank, CBQ, QIB) in QAR
Location / GPS	React Native Geolocation + Google Maps Platform for support worker check-in/check-out and visit-location verification
File Storage	AWS S3 for document uploads, evidence photos, session recordings
Notifications	Firebase Cloud Messaging (mobile push), Nodemailer (email), in-app alerts
Hosting	AWS EC2 (backend), Vercel (web frontend), Expo EAS (mobile build & OTA updates)

New in this version:

- **Payments** — local Qatari bank / NAPS-linked payment gateway
- **Location / GPS** — real-time worker location tracking on mobile

Removed from this version: AI/LLM integration layer (OpenAI GPT-4o) for intake summarisation, care plan drafting, and risk detection.

2. Platform Roles & Access

The platform enforces strict role-based access control (RBAC). Each role has a distinct UI shell, permission set, and workflow scope. All role transitions and permission overrides are logged to the immutable audit trail.

Role	Key Responsibilities & Permissions
Family	Onboarding submission, care plan review & approval, session booking, invoice viewing & online payment, telehealth join, secure messaging
Patient	Limited care info view, telehealth participation only
Care Coordinator	Full care episode lifecycle management, intake review, care plan building, staff assignment, scheduling, family communication, monitoring
Clinical Staff	Assessments, clinical notes, session outcomes, care plan read/limited write, telehealth conduct
Support Worker	Daily schedule view, GPS check-in/check-out at visit location, visit notes, task checklist completion, evidence upload
Admin	Staff management, system configuration, pricing management, payment reconciliation, reporting dashboards, compliance settings

The AI System role from the previous version has been removed; all intake review and care-plan drafting decisions are performed directly by human staff.

3. Milestone Breakdown

The project is structured into 5 milestones across 6 weeks. Each milestone delivers a working, testable increment of the platform. Payment is tied to milestone delivery and approval.

Milestone	Focus	Weeks	Payment
M1 — Foundation & Auth	Infrastructure, DB schema, auth, RBAC, organization setup	Week 1	$500
M2 — Onboarding & Patient Management	Family onboarding, patient profiles, care episodes, assessments	Week 2	$600
M3 — Care Planning & Clinical Ops	Care plan builder, scheduling, clinical notes, care delivery	Weeks 3–4	$700
M4 — Telehealth, Payments & Comms	Video sessions, Qatari bank payment gateway, billing/invoicing, messaging, notifications	Week 5	$600
M5 — Mobile App, GPS, Reporting & Launch	React Native app, GPS worker tracking, admin reporting, audit logs, QA, deployment	Week 6	$600

TOTAL — Full platform delivery — 6 Weeks — $3,000

Milestone 1 — Foundation & Infrastructure (Week 1) | $500

This milestone establishes the entire technical foundation. No clinical features are built until the architecture, database, and authentication are rock-solid.

Backend Setup

- Node.js + Express.js project scaffolding with modular domain-based folder structure (auth, users, organizations, patients, episodes, plans, scheduling, billing, payments, telehealth, audit, notifications)
- PostgreSQL database provisioned on AWS RDS with environment-specific configs (dev/staging/prod)
- Prisma ORM configured with full schema: Users, Roles, Organizations, Clinics, Departments, Employees, CareUnits, Patients, CareEpisodes, Assessments, CarePlans, Goals, Services, Sessions, Invoices, Payments, AuditLogs, Messages, Notifications
- Centralized error handling, request validation (Zod), structured logging (Winston + CloudWatch)
- Environment configuration management with secrets stored in AWS Secrets Manager

Authentication & RBAC

- JWT-based authentication with access token (15min) + refresh token (7-day, rotating)
- bcrypt password hashing (salt rounds: 12) for all user credentials
- RBAC middleware enforced at route level for all 6 user roles
- Session management, forced logout, device token management
- Password reset via time-limited OTP (email, 10-min expiry)
- API rate limiting per user/IP

Milestone 1 Deliverables

- Fully functional authentication system (web)
- PostgreSQL database with complete schema and migrations
- RBAC middleware protecting all routes
- Org hierarchy management (Admin)
- Adaptive UI shell with role-based navigation
- Audit logging foundation active across all endpoints

- Deployed to staging environment (EC2 + Vercel)

Milestone 2 — Onboarding, Patient Management & Assessments (Week 2) | $600

This milestone delivers the intake and onboarding workflow end-to-end — the entry point for every patient journey.

Family Onboarding & Intake

- Multi-step onboarding form: Care Unit creation, Patient profile, primary caregiver details, reason for care, medical history, current medications, diagnoses, urgency level, service interest selection
- Document upload at intake: referral letters, prior assessments, insurance documents (AWS S3, signed URL uploads)
- Structured intake record: immutable, timestamped, linked to family account
- Intake status tracking: Submitted → Under Review → Validated → Episode Created
- Care Coordinator dashboard: queue of new onboarding submissions, filters by status/urgency, intake detail view

Coordinator-Led Intake Review

- Coordinator manually reviews each intake submission and produces a structured clinical summary directly within the platform (key concerns, urgency indicators, suggested service categories, risk flags)
- Summary form uses guided structured fields rather than free text to keep data consistent
- Every review action is logged in AuditLogs with reviewer identity, timestamp, and final summary

Patient & Care Unit Management, Care Episode Management, Assessment Management — unchanged from the original scope: structured patient/care unit profiles, versioned edit history, multi-episode support, and full assessment scheduling/booking/reporting lifecycle.

Milestone 2 Deliverables

- Complete multi-step family onboarding flow (web)
- Structured coordinator-led intake review workflow
- Patient and Care Unit profile management
- Care Episode creation and lifecycle management
- Assessment scheduling, booking, and reporting system
- Document upload and management (S3-backed)
- Coordinator onboarding queue and review dashboard

Milestone 3 — Care Planning, Scheduling & Care Delivery (Weeks 3–4) | $700

This is the most clinically complex milestone. It delivers the care plan builder, the full approval workflow, staff scheduling, care delivery tracking, and clinical documentation — the operational heart of the platform.

Care Plan Builder

- Care Coordinator builds the Care Plan directly within the Care Episode (post-assessment) using a structured template: Goals (SMART format), recommended Services (type, frequency, duration), assigned staff roles, notes
- Reusable goal/service templates speed up plan creation for common episode types
- Care Plan stored as a structured object: Goals[] with title, description, target date, measurement criteria, status; Services[] with type, frequency, unit cost, assigned staff
- Version control: every edit creates a new plan version, prior versions immutable

Care Plan Approval Workflow

- Coordinator submits Care Plan for family approval — notification sent (email + in-app)

- Family reviews Care Plan via portal: goals, services, schedule, cost breakdown
- Family can Approve, Request Changes (with comments), or Reject
- Clinical staff can co-sign care plans within their scope
- On approval, status changes to Active and goals convert into trackable tasks
- Re-approval required for any material change to an active Care Plan

Staff Scheduling & Availability Engine, Care Delivery, Clinical Documentation, Goal & Progress Tracking — unchanged from the original scope: recurring availability, slot-based booking with conflict detection, task-based in-person delivery workflow with evidence upload, immutable versioned clinical notes, and goal progress tracking.

Milestone 3 Deliverables

- Structured care plan builder with full editing and version control
- Family care plan approval/rejection workflow
- Staff availability management and slot-based scheduling engine
- Recurring appointment generation with conflict detection
- In-person care delivery task workflows for Support Workers
- Structured clinical documentation system with immutable history
- Goal and progress tracking linked to Care Plan

Milestone 4 — Telehealth, Payments & Communication (Week 5) | $600

This milestone integrates telehealth as a primary care delivery channel, builds the full billing/invoicing system with local Qatari bank payment collection, and delivers secure multi-party messaging and notifications.

Telehealth Integration

- Daily.co (or Twilio Video) SDK integrated for browser and React Native video sessions
- Session creation workflow: appointment confirmed → telehealth link generated → notification sent
- Waiting room: patient enters on joining; clinical staff admits when ready
- Shared clinical context view during session: active Care Plan, goals, and prior notes in a side panel
- Session notes created during or immediately post-call
- Optional session recording, stored in S3 with access controls

Billing & Payments (Qatar Local Banks)

- Service pricing master: Admin configures pricing per service type — backend-calculated, no client-side price manipulation possible
- Automated quotation generation: itemized service quotation on Care Plan approval, in QAR
- Quotation PDF export: itemized breakdown (service name, rate, quantity, subtotal, tax, total)
- Invoice generation: on session/service completion, invoice line items auto-generated
- **Payment gateway integration** with a locally licensed Qatari payment provider (e.g. PayTabs / Fatora / Skipcash), supporting NAPS-linked debit cards and major local bank cards (QNB, Doha Bank, Commercial Bank of Qatar, Qatar Islamic Bank) for direct in-app / in-portal payment of invoices
- Payment webhook handling for real-time payment status updates (Pending, Successful, Failed, Refunded)
- Invoice lifecycle: Draft → Sent → Paid → Overdue → Disputed, with payment method and transaction reference stored against each paid invoice
- Admin payment reconciliation dashboard: matches gateway settlement reports against recorded invoices
- Family can view invoices/quotations and pay directly via the portal or mobile app

Communication — Secure Messaging, Notifications — unchanged from the original scope: context-aware messaging threads linked to Care Episodes, real-time updates via Socket.io, and full in-app/email/push notification coverage — now including payment-related events (payment received, payment failed, invoice overdue).

Milestone 4 Deliverables

- Fully functional telehealth video sessions (web + mobile)
- Waiting room, session controls, shared care plan view during calls
- Service pricing configuration and automated quotation generation (QAR)
- Live Qatari bank / NAPS payment gateway integration for invoice payment
- Invoice generation, lifecycle management, payment reconciliation, and PDF export
- Secure multi-party messaging with real-time updates
- Full notification system: in-app, email, and mobile push

Milestone 5 — React Native Mobile App, GPS Tracking, Reporting & Deployment (Week 6) | $600

The final milestone delivers the complete React Native mobile application including GPS-based worker location tracking, the Admin reporting and analytics dashboards, the audit and compliance module, full QA testing, and production deployment.

React Native Mobile Application

- Built with React Native (Expo managed workflow) for iOS and Android from a single codebase

- NativeWind for consistent styling with web; React Navigation v6; Zustand for state management
- Family role app screens: Onboarding, Home dashboard, Care Episode view, Care Plan review & approval, Appointment booking, Telehealth join, Messaging, Invoice viewer & in-app payment, Notification center
- Clinical Staff role app screens: Patient list, Daily schedule, Assessment completion, Session notes entry, Telehealth join, Care Plan view
- Support Worker role app screens: Daily schedule, GPS check-in/check-out, Task checklist, Note entry, Evidence photo upload
- Push notification integration via Firebase Cloud Messaging
- Offline capability for critical read-only views (care plan, schedule) with sync on reconnect
- Expo EAS build pipeline configured for TestFlight (iOS) and Google Play internal track (Android)

GPS Worker Location Tracking

- Support Worker mobile app requests location permission and captures GPS coordinates at visit check-in and check-out
- Geofencing validation: check-in location compared against the patient's registered address, with a configurable acceptable-radius tolerance
- Live location snapshot stored against the session record (latitude, longitude, timestamp, accuracy) for audit and compliance purposes
- Coordinator dashboard map view showing active field workers and their last known check-in location for the day
- Missed or out-of-radius check-ins flagged automatically to the Care Coordinator
- All location captures are logged in the Audit trail alongside the associated session and worker identity

Admin Reporting & Analytics Dashboard

- Care Outcomes Dashboard: active episodes, goals achieved vs. in-progress, episode closure rate, average episode duration
- Service Utilisation Reports: sessions delivered per service type, staff utilisation, telehealth vs in-person split
- Staff Performance Metrics: sessions completed, documentation turnaround time, task completion rates, GPS check-in compliance rate
- Financial Reporting: total revenue by period, invoiced vs. collected via payment gateway, outstanding amounts
- All reports exportable as CSV and PDF, with date range, clinic/department, and staff/patient filtering

Audit & Compliance Module

- Audit log viewer (Admin only): searchable, filterable log of every system action
- Care Plan edit history viewer with diff view between versions
- Payment transaction log: every gateway transaction with status, method, and reference number
- GPS check-in/check-out log viewer with map replay per session
- Data export for compliance: audit logs exportable in CSV and JSON
- Data retention policy configuration; GDPR/HIPAA-aligned patient data deletion workflow

QA & Production Deployment — unchanged from the original scope: end-to-end testing of all role journeys, API integration testing, cross-browser and mobile device testing, security and performance testing (including payment gateway sandbox-to-live cutover testing), and deployment to AWS EC2 / RDS, Vercel, and Expo EAS, with full handover documentation.

Milestone 5 Deliverables

- Complete React Native mobile app (iOS + Android) for Family, Clinical Staff, Support Worker roles
- GPS-based worker check-in/check-out with geofencing and compliance reporting
- Admin analytics and reporting dashboards with export capability

- Full audit log viewer and compliance module, including payment and location logs
- End-to-end QA across all user roles and journeys
- Production deployment on AWS + Vercel + Expo EAS
- Full handover documentation and walkthrough

4. Investment & Payment Schedule

Milestone	Deliverable Focus	Weeks	Payment	% Total
M1 — Foundation	Infrastructure, DB, Auth, RBAC, Org Setup	1	$500	17%
M2 — Onboarding	Intake, Structured Review, Patients, Episodes, Assessments	2	$600	20%
M3 — Care Planning	Care Plan Builder, Approval, Scheduling, Delivery, Notes	3–4	$700	23%
M4 — Telehealth & Payments	Video Sessions, Qatari Bank Payment Gateway, Invoicing, Messaging	5	$600	20%
M5 — Mobile, GPS & Launch	React Native App, GPS Tracking, Reporting, Audit, QA, Deploy	6	$600	20%
TOTAL	**Full Platform Delivery**	**6 Weeks**	**$3,000**	**100%**

Payment is due at the start of each milestone upon client approval of the prior milestone's deliverables. Third-party service costs (AWS, Daily.co/Twilio, Firebase, and the Qatari payment gateway's own transaction/processing fees) are billed separately to the client's own accounts.

5. Scope Inclusions & Exclusions

5.1 Included in Scope

- All 6 user roles with full RBAC implementation
- Web application (Next.js) — all role dashboards and workflows
- React Native mobile app — Family, Clinical Staff, Support Worker roles
- PostgreSQL database with full relational schema and Prisma ORM
- Telehealth video sessions (Daily.co or Twilio Video)
- Local Qatari bank / NAPS payment gateway integration for invoice and quotation payment
- GPS-based worker location tracking with geofencing and compliance reporting
- Billing, quotation generation, and invoice management
- Secure messaging and real-time notifications
- Audit trail across all system actions
- Admin reporting and analytics dashboards
- Deployment to AWS (backend), Vercel (web), Expo EAS (mobile)
- Full handover documentation and walkthrough session

5.2 Excluded from Scope

- AI/LLM-based intake summarisation, care plan drafting, or risk detection — removed from this version
- Multi-language / localization support
- Native iOS/Android public app store listing (TestFlight and Play internal track included; public store listing is client's responsibility)
- Ongoing maintenance and hosting post-delivery

- Third-party API subscription and transaction costs (AWS, Daily.co/Twilio, Firebase, payment gateway processing fees — all client-side accounts)
- Background/continuous GPS tracking outside of active check-in/check-out sessions (privacy-by-design: location is only captured at visit start/end, not continuously)

6. Terms & Working Agreement

Communication	Weekly progress update every Monday. Async via WhatsApp/Slack; video calls at each milestone review.
Revisions	Up to 2 revision rounds per milestone. Additional revisions billed at $40/hr.
Feedback SLA	Client provides feedback within 3 business days of milestone delivery. Delays may extend timeline.
Source Code	Full source code ownership transferred to client upon final payment. GitHub repo access provided from day 1.
Confidentiality	All patient data, payment data, system design, and business logic treated as strictly confidential. NDA available upon request.
Scope Changes	New features outside this proposal's scope are assessed and quoted separately. No scope creep without written agreement.

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